Paired-Ridge Learning Experiment

Question and design. This experiment asks whether the broker's ranking advantage persists when both agents and the broker use Ridge regression instead of neural networks. Agent models predict match quality from the candidate's type. The broker predicts match quality from the pair's additive and bilinear type features. Thus, both learners use linear Ridge estimators, with separately calibrated penalties, while the broker retains the pair representation and pooled history made possible by mediation. The experiment therefore tests whether neural-network flexibility is necessary for a broker ranking advantage. It does not equalize the learners' information sets or feature maps. The calibrated penalties are $\lambda_a = 0.003$ and $\lambda_b = 0.001$.

The primary estimand in realization c is

$$G_L(c) = \overline{r}_{b,L}(c) - \overline{r}_{a,L}(c),$$

where $L \in \{\mathrm{NN}, \mathrm{Ridge}\}$ and each bar is the seed-averaged holdout rank correlation over periods 481–500. The learning-system contrast is $G_{\mathrm{Ridge}}(c) - G_{\mathrm{NN}}(c)$. It includes the endogenous changes in matching and subsequent learning histories induced by the learning rule; it is not a fixed-data comparison of estimators. Each of the 98 effective realizations receives one observation. Grid coordinates that refer to the same realization receive no additional weight. The model comparison uses the same seeds in both learning conditions: 50 at the baseline and 20 elsewhere.

Figures R1–R4 reproduce the full content of Main Figures 1–4 with NN and paired Ridge shown in the same figure. Corresponding panels use shared axes, so differences in level and dispersion are visually comparable. Figure R5 then shows direct paired comparisons of the primary ranking and output outcomes.

Baseline. With the common seeds, the NN broker-agent rank-correlation gap is 0.392, compared with 0.335 under paired Ridge. The paired change is -0.057, with a 95% Monte Carlo interval of [-0.082, -0.032]. Table R1 reports the other baseline outcomes. Figure R1 shows how the baseline's matching, network, access, and outsourcing dynamics differ over the full run. Figure R5A isolates the corresponding broker and agent ranking trajectories.

Robustness across model realizations. Across the 98 effective realizations, the median broker-agent rank-correlation gap is 0.376 for NN and 0.342 for Ridge. The median Ridge-minus-NN contrast is +0.005. Its 10th and 90th percentiles are -0.085 and +0.205. The Ridge gap is positive in 98 realizations, compared with 96 under NN, and exceeds the NN gap in 51 realizations. The correlation between NN and Ridge condition-level rank gaps is 0.670.

The within-system brokered-minus-self output gap has medians of 0.516 under NN and 0.458 under Ridge. Its median Ridge-minus-NN contrast is -0.095, and Ridge exceeds NN in 11 realizations. This channel mean difference is descriptive, not a causal estimate of the value of brokerage: channel choice, candidate sets, and offer acceptance are endogenous. Median outsourcing is 0.961 under NN and 0.928 under Ridge.

Table R1: Baseline late-period outcomes under NN and paired-Ridge learning

Outcome	NN	Ridge	Ridge − NN
Agent holdout rank correlation	0.503	0.479	-0.024
Broker holdout rank correlation	0.895	0.814	-0.081
Broker-agent rank-correlation gap	0.392	0.335	-0.057
Prediction R^2 gap	3.790	2.825	-0.964
Brokered-minus-self output gap	0.652	0.530	-0.121
Outsourcing rate	0.979	0.939	-0.040
Broker access fraction	0.109	0.114	+0.005
Broker betweenness centrality	0.825	0.783	-0.043

Entries use the 50 common seeds. Each seed is first averaged over periods 481–500. The Ridge-minus-NN column is the mean paired seed difference. The access fraction is the share of accepted broker offers whose counterparty was not already the demander's neighbor.

Figure R2 shows the matching-problem surface under each learning system. Figure R3 compares the broker's evolving network position and access, and Figure R4 shows how the rank and output gaps covary with those structural outcomes across effective realizations. Figure R5B–C summarizes the paired condition-level differences directly against zero.

Interpretation. The paired-Ridge results show that neural-network flexibility is not necessary for the broker to rank candidates better than agents in this modeled system: the Ridge gap is positive in 98 of 98 effective realizations. This result does not identify the cause of that advantage. The broker still observes both parties to each mediated match, represents the pair with additive and bilinear features, and pools observations across mediated matches. The Ridge broker ablations probe the system-level consequences of restricting pooled sample size, retaining only one endpoint type per fitting row, and removing bilinear pair features.

Illustrative figures

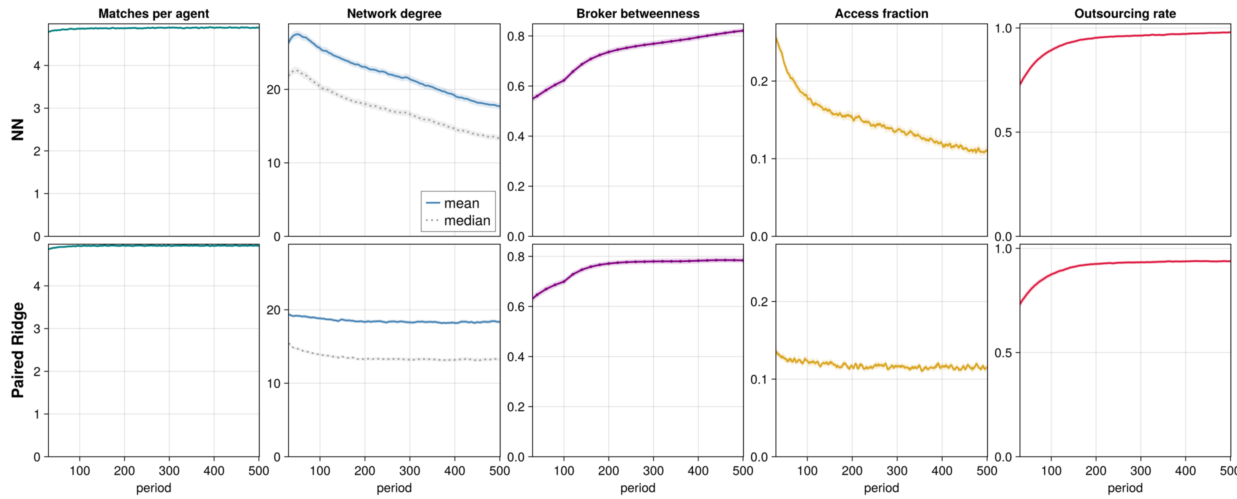

Figure R1: Baseline dynamics under NN and paired Ridge. Rows identify the learning system. Panels show matches per agent, mean and median agent-network degree, broker betweenness centrality, access fraction, and outsourcing rate. Lines are five-observation rolling ensemble means over 50 common seeds. Ribbons are pointwise 95% Monte Carlo intervals. Betweenness is measured every 20 periods; other measures are recorded every period. Axes begin at period 30 and are shared across learning systems.

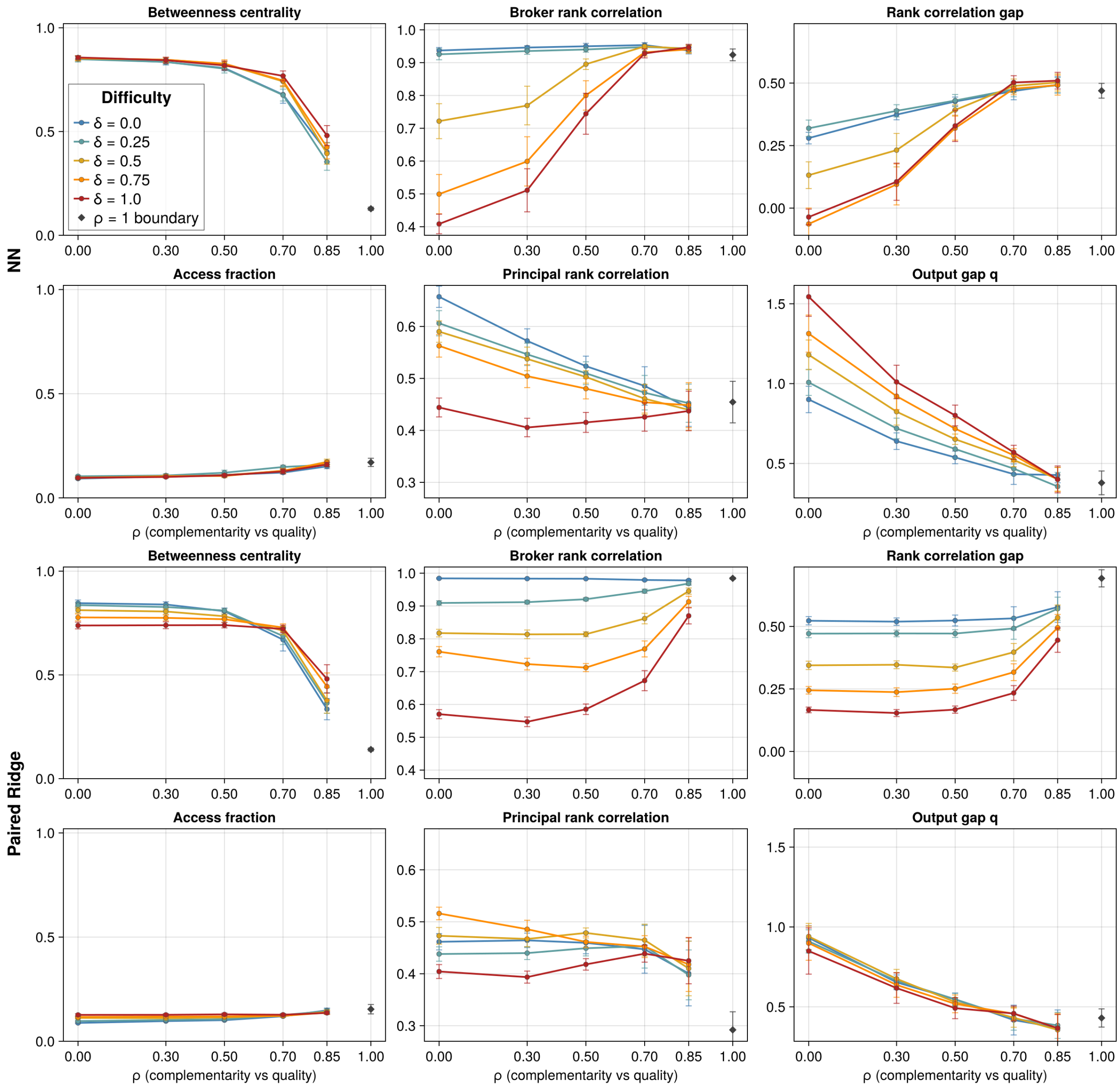

Figure R2: Outcomes across the $\rho \times \delta$ matching grid under NN (top two rows) and paired Ridge (bottom two rows). Each line is a value of δ; lines span $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Each marker is an effective realization's late-window seed mean. Corresponding NN and Ridge panels use the same vertical scale. Whiskers are 95% Monte Carlo intervals.

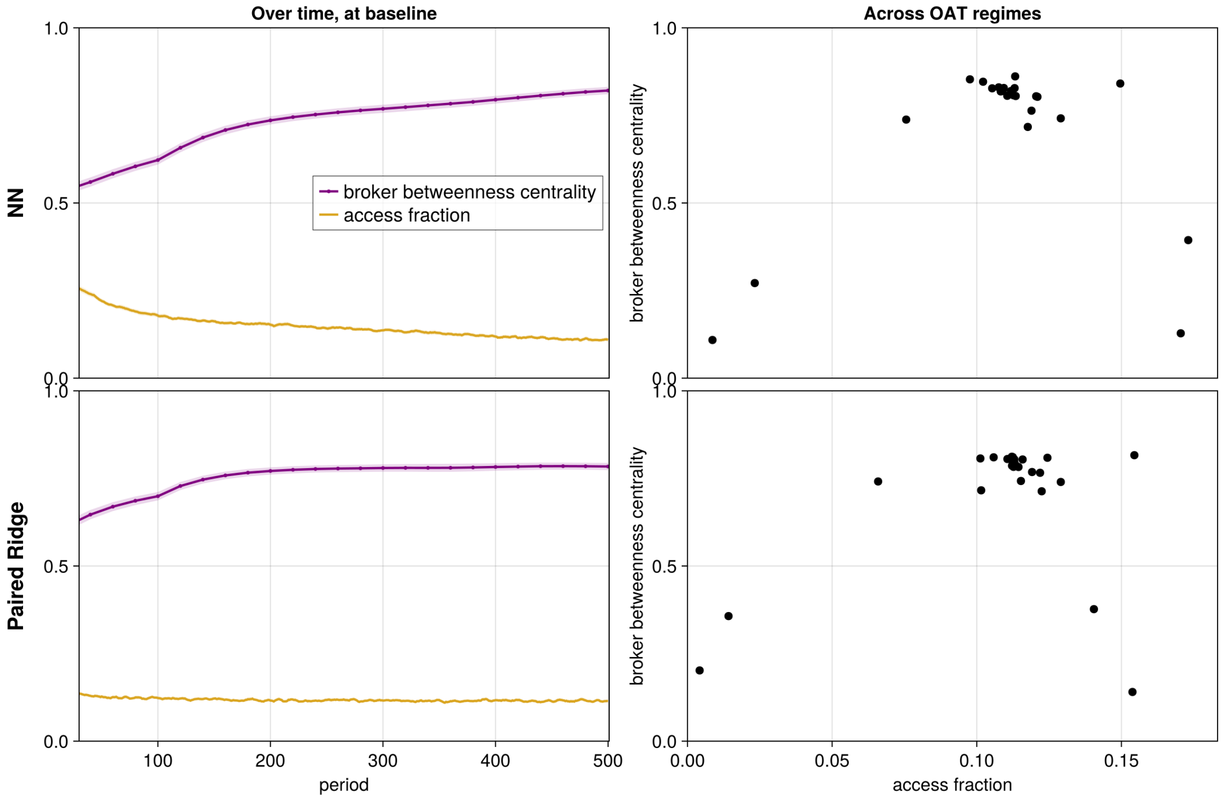

Figure R3: Broker position and access under NN and paired Ridge. Left panels show five-observation rolling ensemble means at the baseline. Right panels show one point per one-at-a-time regime, using late-window seed means. Ribbons are pointwise 95% Monte Carlo intervals. Corresponding panels use shared axes.

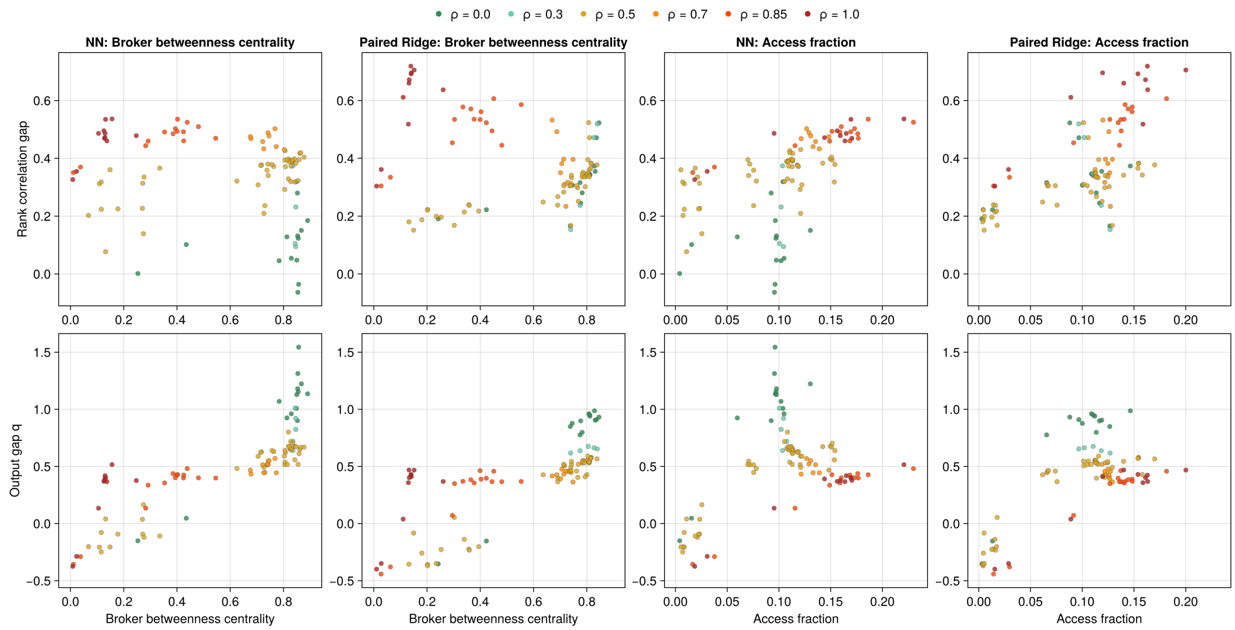

Figure R4: Rank-correlation and brokered-minus-self output gaps against broker betweenness centrality and access fraction under NN and paired Ridge. Each point is one effective realization's late-window seed mean. Corresponding NN and Ridge panels use shared axes; color denotes ρ.

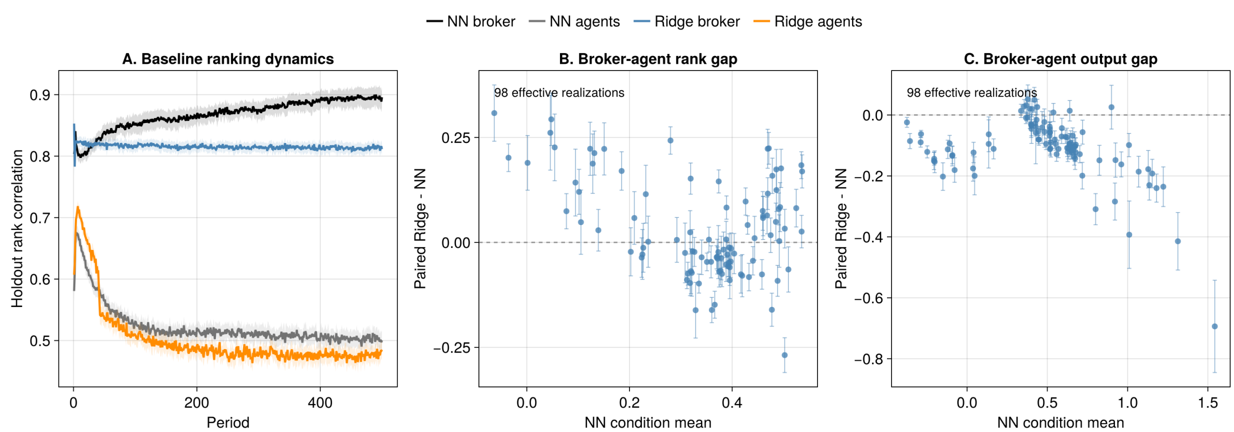

Figure R5: Direct paired-Ridge comparison. Panel A uses the 50 common baseline seeds, with pointwise 95% Monte Carlo intervals. Panels B and C plot the paired Ridge-minus-NN difference for each of the 98 effective realizations against that realization's NN mean. Whiskers are 95% intervals for the mean seed-level difference, and dashed lines mark no difference.